

TIRA E-COMMERCE DATABASE MANAGEMENT SYSTEM

PAYEMENT MANAGEMENT SYSTEM

1.Database Design

1.1Payment Table

Attribute	Data Type	Constraint
Payment_ID	NUMBER	Primary Key
Order_ID	NUMBER	Foreign Key
Payment_Mode	VARCHAR2(20)	NOT NULL
Payment_Date	DATE	NOT NULL
Payment_Amount	NUMBER(10,2)	NOT NULL
Payment_Status	VARCHAR2(20)	NOT NULL

2.Table Relationships

Customer → Orders

One customer can place many orders.

Orders → Payment

Each order is associated with a payment transaction.

Orders → Order_Details

One order can contain multiple products.

3.Table Creation

3.1Create Payment Table

```
SQL> CREATE TABLE Payment (  
2     Payment_ID NUMBER PRIMARY KEY,  
3     Order_ID NUMBER,  
4     Payment_Mode VARCHAR2(20) NOT NULL,  
5     Payment_Date DATE NOT NULL,  
6     Payment_Amount NUMBER(10,2) NOT NULL,  
7     Payment_Status VARCHAR2(20) NOT NULL,  
8     FOREIGN KEY (Order_ID) REFERENCES Orders(Order_ID)  
9 );
```

Table created.

4.Payment Data Insertion

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)
  3 VALUES
  4 (501, 1001, 'UPI', SYSDATE, 700.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)
  3 VALUES
  4 (502, 1002, 'Card', SYSDATE, 1200.00, 'Successful');
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)
  3 VALUES
  4 (503, 1003, 'Net Banking', SYSDATE, 850.00, 'Failed');
```

1 row created.

```
SQL> INSERT INTO Payment
  2 (Payment_ID, Order_ID, Payment_Mode, Payment_Date, Payment_Amount, Payment_Status)
  3 VALUES
  4 (504, 1004, 'UPI', SYSDATE, 1500.00, 'Successful');
```

1 row created.

5. Manage Payment Transactions

5.1 Display Successful Payments

```
SQL> SELECT *  
  2 FROM Payment  
  3 WHERE Payment_Status = 'Successful';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
501	1001	UPI	28-SEP-26	700
502	1002	Card	28-SEP-26	1200
504	1004	UPI	28-SEP-26	1500

5.2 Display Failed Payments

```
SQL> SELECT *  
  2 FROM Payment  
  3 WHERE Payment_Status = 'Failed';
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	1003	Net Banking	28-SEP-26	850

6.Modify Payment Status

```
SQL> UPDATE Payment
  2 SET Payment_Status = 'Successful'
  3 WHERE Payment_ID = 503;
```

1 row updated.

```
SQL> SELECT *
  2 FROM Payment
  3 WHERE Payment_ID = 503;
```

PAYMENT_ID	ORDER_ID	PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT
503	1003	Net Banking	28-SEP-26	850
Successful				

7.Analyze Payment Methods

```
SQL> SELECT
  2     Payment_Mode,
  3     COUNT(*) AS Total_Transactions
  4 FROM Payment
  5 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_TRANSACTIONS
UPI	2
Card	1
Net Banking	1

Calculate Amount Collected By Payment Mode

```
SQL> SELECT
  2     Payment_Mode,
  3     SUM(Payment_Amount) AS Total_Amount
  4 FROM Payment
  5 GROUP BY Payment_Mode;
```

PAYMENT_MODE	TOTAL_AMOUNT
UPI	2200
Card	1200
Net Banking	850

8.Payment Transaction Report

```
SQL> SELECT
  2     p.Payment_ID,
  3     p.Order_ID,
  4     c.Customer_ID,
  5     c.First_Name || ' ' || c.Last_Name AS Customer_Name,
  6     p.Payment_Mode,
  7     p.Payment_Date,
  8     p.Payment_Amount,
  9     p.Payment_Status
 10 FROM Payment p
 11 JOIN Orders o
 12     ON p.Order_ID = o.Order_ID
 13 JOIN Customer c
 14     ON o.Customer_ID = c.Customer_ID
 15 ORDER BY p.Payment_Date DESC;
```

PAYMENT_ID	ORDER_ID	CUSTOMER_ID
------------	----------	-------------

CUSTOMER_NAME

PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
--------------	-----------	----------------	----------------

504	1004	104	
-----	------	-----	--

Kavin M			
---------	--	--	--

UPI	28-SEP-26	1500	Successful
-----	-----------	------	------------

503	1003	103	
-----	------	-----	--

Anitha R			
----------	--	--	--

Net Banking	28-SEP-26	850	Successful
-------------	-----------	-----	------------

PAYMENT_ID	ORDER_ID	CUSTOMER_ID
------------	----------	-------------

CUSTOMER_NAME

PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
--------------	-----------	----------------	----------------

502	1002	102	
Priya S			
Card	28-SEP-26	1200	Successful

501	1001	101	
Harini N			

PAYMENT_ID	ORDER_ID	CUSTOMER_ID
------------	----------	-------------

CUSTOMER_NAME

PAYMENT_MODE	PAYMENT_D	PAYMENT_AMOUNT	PAYMENT_STATUS
--------------	-----------	----------------	----------------

UPI	28-SEP-26	700	Successful
-----	-----------	-----	------------